

CHAPTER 5

Cancer diagnostics and treatment decisions using artificial intelligence

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Abstract

Recent advances in computational science mean that the capacity of artificial intelligence (AI) to predict disease risk, diagnose illness, and guide therapeutic decision making is now a realistic prospect in the foreseeable future. The use of AI has already shown promise in a wide variety of fields in medicine including cardiology, renal medicine, critical care, and mental health. For a variety of reasons, cancer is an area where AI is expected to have a significant impact in the near term, and cancer is in many ways an obvious choice for enhancing the “reputation” of AI-based approaches in healthcare. First, cancer is common and imposes a considerable burden in terms of physical disability, emotional trauma, and economic cost for patients and society as a whole. Thus, almost without exception, across all cancer subtypes, there is a critical need to improve cancer-related outcomes. Second, at the individual level, there is huge patient demand for enhanced cancer diagnostics, prognostication, and therapy; cancers represent a major and indiscriminate cause of morbidity and mortality, and they evoke fear in a way that few other conditions do, because of the perception of cancer as a fatal diagnosis. From the cancer clinician’s perspective, there is an equally urgent drive since tumor heterogeneity and the exponential growth in onco-therapeutic options present major challenges in terms of decision making, not least due to sheer data burden. Rather alarmingly, it has been estimated that the modern-day cancer physician would need to spend in excess of 20 hours per day reading, in order to stay up to date with developments in the scientific literature. This challenge is well exemplified by current attempts to unravel the cancer genome—millions of molecular disparities are now coming to light that could, alone or in combination, influence cancer cell survival and progression, but harnessing these discoveries in a meaningful way is impossible without state-of-the-art computing. The impact of AI approaches is also expected to be significant for academia, where cancer research initiatives are likely to be more precisely steered by AI-based data-mining discoveries. This in turn is likely to mean that certain avenues of research in terms of cancer diagnostics and therapy are “shelved” either permanently or temporarily until new developments emerge that suggest otherwise. Furthermore, it is anticipated that for the pharmaceuticals industry, anticancer drug development and the development of companion diagnostics will benefit from an additional layer of premarket delivery validation, through AI-based analyses of actual and simulated clinical outcomes. One can thus appreciate that all stakeholders within the cancer health space stand to make potentially significant gains with the application of AI solutions.

Keywords

Artificial intelligence; machine learning; deep learning; cancer; artificial neural network; next-generation diagnostics

5.1 Background

Recent advances in computational science mean that the capacity of artificial intelligence (AI) to predict disease risk, diagnose illness, and guide therapeutic decision making is now a realistic prospect in the foreseeable future [1]. The use of AI has already shown promise in a wide variety of fields in medicine including cardiology, renal medicine, critical care, and mental health [2–5]. For a variety of reasons, cancer is an area where AI is expected to have a significant impact in the near term, and cancer is in many ways an obvious choice for enhancing the “reputation” of AI-based approaches in healthcare. First, cancer is common and imposes a considerable burden in terms of physical disability, emotional trauma, and economic cost for patients and society as a whole. Thus, almost without exception, across all cancer subtypes, there is a critical need to improve cancer-related outcomes. Second, at the individual level, there is huge patient demand for enhanced cancer diagnostics, prognostication, and therapy; cancers represent a major and indiscriminate cause of morbidity and mortality, and they evoke fear in a way that few other conditions do, because of the perception of cancer as a fatal diagnosis. From the cancer clinician’s perspective, there is an equally urgent drive since tumor heterogeneity and the exponential growth in onco-therapeutic options present major challenges in terms of decision making, not least due to sheer data burden. Rather alarmingly, it has been estimated that the modern-day cancer physician would need to spend in excess of 20 hours per day reading, in order to stay up to date with developments in the scientific literature. This challenge is well exemplified by current attempts to unravel the cancer genome—millions of molecular disparities are now coming to light that could, alone or in combination, influence cancer cell survival and progression, but harnessing these discoveries in a meaningful way is impossible without state-of-the-art computing. The impact of AI approaches is also expected to be significant for academia, where cancer research initiatives are likely to be more precisely steered by AI-based data-mining discoveries. This in turn is likely to mean that certain avenues of research in terms of cancer diagnostics and therapy are “shelved” either permanently or temporarily until new developments emerge that suggest otherwise. Furthermore, it is anticipated that for the pharmaceuticals industry, anticancer drug development and the development of companion diagnostics will benefit from an additional layer of premarket delivery validation, through AI-based analyses of actual and simulated clinical outcomes. One can thus appreciate that all stakeholders within the cancer health space stand to make potentially significant gains with the application of AI solutions (Fig. 5.1).

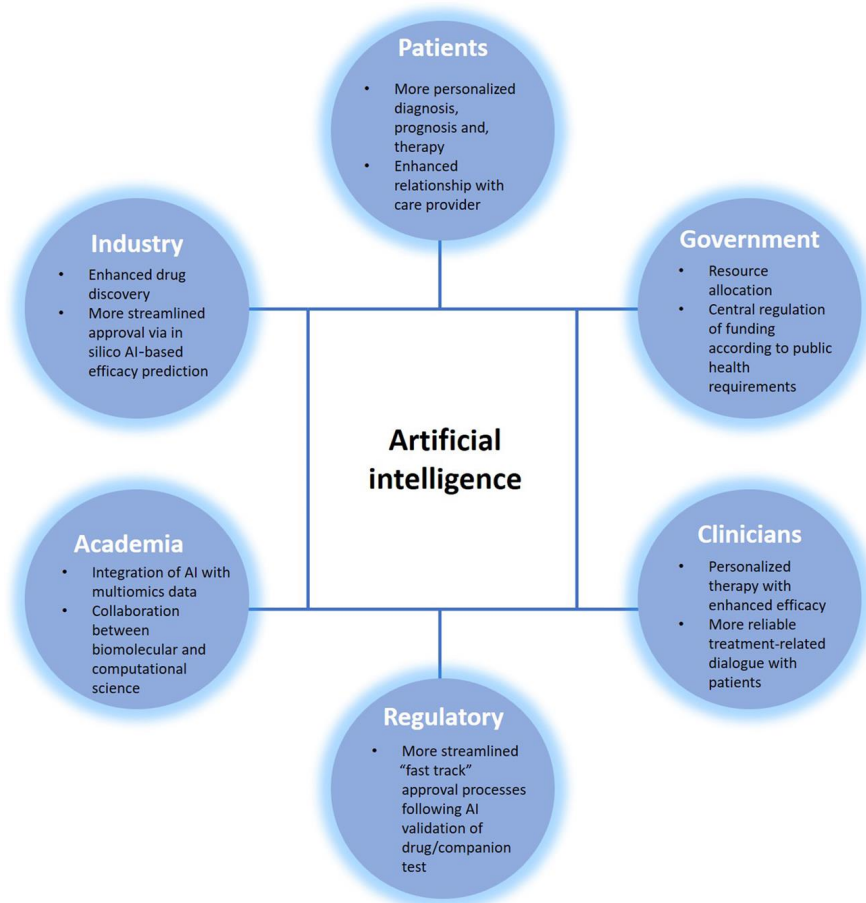


FIGURE 5.1 Expected impact of AI-based approaches on different stakeholder groups in the cancer health space.

To better understand where the technology is headed, it is first essential to recognize where we are currently and to define the areas of critical unmet need in cancer medicine. Currently, in the case of solid organ tumors, criteria that determine patient prognosis and management relate to a variety of factors, such as histological tumor grade, tumor stage, and the presence/absence of prognostically relevant features such as vascular and/or lymphatic invasion. More recently, molecular phenotyping data have been successfully incorporated into the patient pathway, notable examples including hormone receptor status in breast cancer and KRAS mutational status in Stage IV colorectal cancer [6,7]. As our understanding of cancer development and progression becomes more sophisticated, so data pertaining to a given patient become ever more multidimensional. The ideal computational pipeline under these circumstances would be capable of seamlessly integrating data from multiple heterogeneous sources (molecular/clinopathological/radiological/exposomal) to arrive at a given answer (risk prediction/cancer diagnosis/cancer prognostication/determination of therapeutic efficacy). Conventional dimensionality reduction statistical approaches of linear regression and logistic regression (LR) have yielded some useful insights into these areas. However, they are now increasingly subject to fundamental limitations imposed by the increasing complexity of biomedical data. In recent years, AI approaches have shown promise in this area as they utilize a variety of statistical, probabilistic, and optimization techniques and parameters to “learn” from past case examples and to detect latent clinically relevant patterns from “noisy”/multidimensional datasets. In this chapter, current and emerging trends in the application of AI approaches in all phases of the cancer patient pathway are summarized.

5.2 Artificial intelligence, machine learning, and deep learning in cancer

The term *artificial intelligence* was first coined by computer scientists in 1956 and now represents a wide umbrella term encompassing a growing selection of algo-

rhythmic disciplines and subdisciplines [8]. Fig. 5.2 provides a graphical summary of research trends over the past two decades and illustrates the steady growth of the AI sector in cancer research. Presently, the terms AI, machine learning (ML), and deep learning (DL) are used somewhat interchangeably in the scientific literature, and to a greater extent in mainstream media. A detailed overview of these computational methods is beyond the scope of this chapter and is provided in [Chapter 2](#), Introduction to Artificial Intelligence. However, it is critical that cancer scientists and cancer physicians have an appreciation of the fundamental differences between these interconnected methods, as they are likely to have impact across different areas. Simplistically, one can consider the key data challenges in cancer medicine as (1) logistical, (2) precision centered, and (3) next generation. It is useful to consider the application of AI, ML, and DL in cancer, across these three domains (Fig. 5.3). From the point of view of *logistics*, there are essential, often labor intensive, human error prone, but relatively basic tasks performed in cancer medicine, where a computational approach that could be trained to perform “as well as” a human counterpart would be highly desirable.

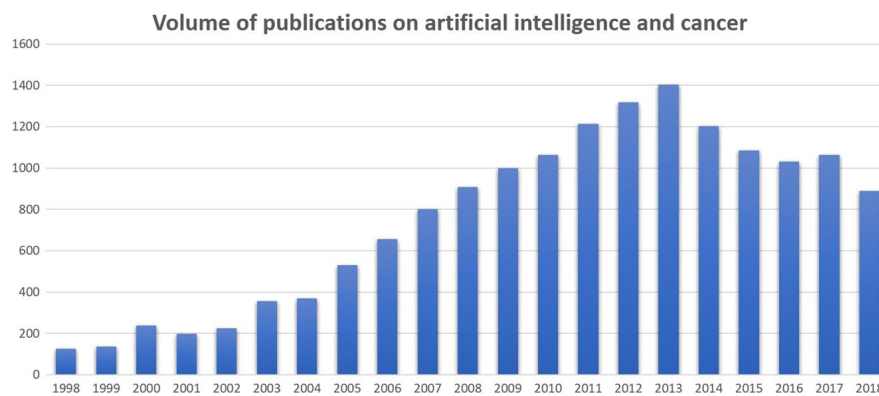


FIGURE 5.2 Graphical representation of the steady growth of AI-based studies in cancer research.

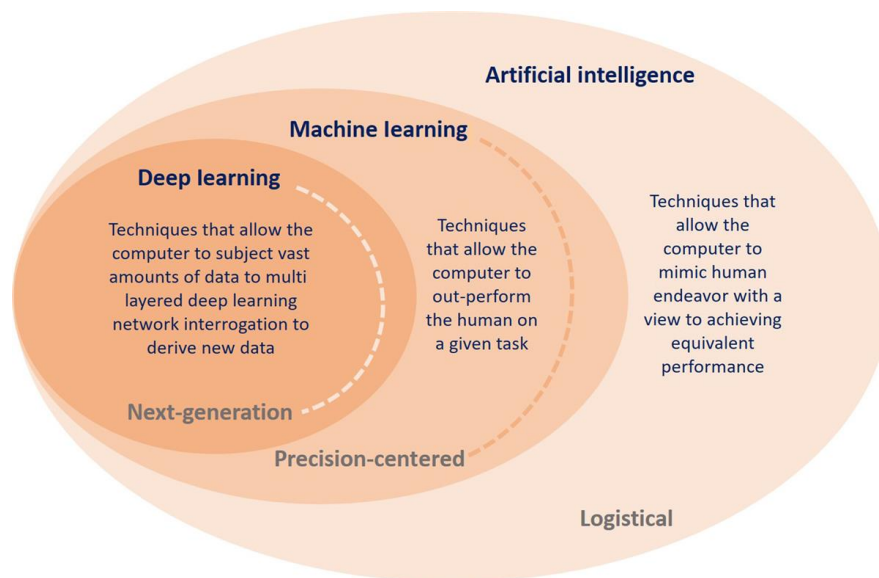


FIGURE 5.3 The role of AGI, ML, and DL in cancer medicine and how these are likely to address logistical, precision-centered, and next-generation objectives in cancer medicine in the future.

This comes down to time efficiency, cost, and logistics of human endeavor in an era where services are attempting to expand, while simultaneously being subjected to an unprecedented level of pressure in terms of available resources. For example, in February 2019 a National Audit Office review reported that more than 150,000 untested cervical screening samples were discovered in laboratories across England [9]. The report concluded that laboratory staff shortages had resulted in a decline in performance against turnaround time targets. In a situation such as this, an AI platform capable of selecting patients for national cervical screening, based on the predefined criteria, and interpreting smear test results as “normal,” “inadequate,” or “abnormal,” before algorithmically defining the neces-

sary follow-up strategy for the patient, would be enormously beneficial. Although this would represent a huge undertaking in practical terms, in computational terms this would constitute artificial general intelligence (AGI). From a *precision-centered* perspective, the requirements are slightly different. Here, the aim is to have a computational pipeline that outperforms a person. For example, oncologists and computer scientists in the Netherlands recently reported that a ML approach was capable of achieving better diagnostic performance than a panel of pathologists for the evaluation of breast cancer lymph node metastases [10]. Here, the algorithm is not coming up with anything “new” or “revolutionary,” and it has been trained by a human to look for the same tissue morphological features that a pathologist would recognize. However, it is seemingly capable of performing the pattern recognition task more consistently and more reliably than the human counterpart. This is ML applied to cancer medicine. In terms of *next-generation* developments, here the goal with an AI approach is to go beyond what is possible with human interpretation and to thereby generate entirely novel knowledge which would otherwise not be derived. This is where DL methods are showing huge promise. DL is more of a computational search of the unknown, potentially revealing latent links and network connections between seemingly disconnected clouds of information. The algorithm is essentially left to its own devices and mines the data using multiple levels of abstraction to learn connections on a deeper level than is capable with the human mind. The program learns through iteratively processing data and requires large amounts of computational power and a database large enough for it to learn. For example, a study published recently in Cancer Research reported on the use of a convolutional neural network (CNN) to evaluate cancer tissue sections and determine the likely radio-sensitivity of the cancer based on a DL approach. The authors found that the way cancer cells clustered together was found by the algorithm to correspond strongly with whether they were radioresistant or radiosensitive [11]. Simplistically, one can consider the key data challenges in cancer medicine as (1) logistical, (2) precision centered, and (3) next generation. It is useful to consider the application of AI, ML, and DL in cancer, across these three domains (Fig. 5.3).

5.3 Artificial intelligence to determine cancer susceptibility

Clinical vignette: Aiko is 38 years of age and has a strong family history of colorectal cancer. At the insistence of her family, she agrees to go and see a gastroenterologist following several episodes of fresh rectal bleeding and mild unintentional weight loss. The colonoscopy demonstrates an impassable malignant-looking stricture in the sigmoid colon, which is subsequently confirmed as adenocarcinoma on histology. Her staging CT scan reveals two metastatic deposits in the liver. She undergoes treatment in the form of systemic chemotherapy, right hemi-hepatectomy, and colorectal resection of her primary tumor. She is now under a strict 6 monthly surveillance program as her oncologist tells her that she is still at risk of cancer relapse, given her advanced stage at presentation. She asks her cancer physician whether more could have been done at an earlier stage to identify and quantify her risk of cancer, given the family history.

The above clinical scenario is a common one in cancer medicine and highlights the critical need for enhanced methods with which to define an individual’s risk of cancer. The principles of P4 medicine applied to cancer were first introduced by Hood and Friend in 2011 [12], describing a model that encapsulates the four key facets of precision healthcare—prediction, prevention, personalization, and participation. Prediction of an individual’s risk of developing a given cancer has the potential to significantly minimize the physical, psychological, and socio-economic burden imposed by cancer. However, at the present time, there is only very limited understanding of the factors that govern cancer susceptibility. It is esti-

mated that only 5%–10% of cancers are the result of inherited genetic mutations [e.g., breast invasive carcinoma 1 (BRCA 1) and 2 in hereditary breast cancer]. On the other hand, approximately 90% of all cancers are thought to develop as a consequence of the accumulation of acquired somatic mutations, triggered by complex interplay between environmental factors, lifestyle factors, and an individual's genomic cancer likelihood. At the present time, large-scale, prospectively maintained, high-quality data on environmental factors, lifestyle behaviors, and dietary practices are lacking, and this is an area that requires urgent attention. However, rapid developments in whole-genome sequencing in recent years have led to the emergence of publicly available genomic data repositories (e.g., the Cancer Genome Atlas), which now offers the potential to be mined using ML approaches for prediction of cancer genomic susceptibility to different cancer types. This potential is exemplified by a recent study by Kim and Kim, who utilized data from The Cancer Genome Atlas (TCGA) database with the aim of developing ML-based genomic models with which to predict risk across 20 major cancer subtypes [13]. The authors specifically sought to address two objectives with this work. First, to estimate the relative proportion of patients with a specific cancer who acquired the condition primarily through inherited genomic susceptibility. In turn this would, by proxy, lead to the indirect identification of patients developing cancer due to uninherited factors (e.g., environment and/or lifestyle). Second, the authors derived estimates of the rank-ordered probability of an individual patient's likelihood of developing one or more cancer phenotypes. The authors evaluated genotype to phenotype association in 5919 individuals with the cohort including 20 different cancer classes and one "healthy" phenotype, and genomics data were entered into a ML data treatment pipeline using a *k*-nearest neighbor method. The authors reported that 33%–88% of cancer cases were associated with a demonstrable inherited genomic abnormality, with the remainder primarily attributed, by corollary, to environmental and lifestyle factors. Importantly, the authors also found that at the individual level they were able to determine the likelihood of developing other linked cancers by evaluating commonality in genomic expression patterns across different cancer subtypes. This work represents a significant breakthrough in terms of the application of AI-based approaches to defining cancer susceptibility. The authors furthermore suggest that information regarding an individual's highest susceptibility to 1 of the 21 phenotypes (20 cancer, 1 healthy) may be useful at the patient-specific level to guide clinician and patient on ways with which to minimize that particular cancer risk through environment and lifestyle modifications (where the inherited genomic susceptibility for cancer is predicted "low"), or to proactively monitor for early detection and intervention of the cancer in individuals with "high" predicted genomic susceptibility.

In a 2016 study, Rau et al. focused on evaluating the association between type II diabetes mellitus (T2DM) and risk of liver cancer, utilizing data from the Taiwan National Health Insurance Research Database subjected to AI-based algorithmic interrogation [14]. A total of 2060 patients were included in the final study and patients were grouped according to whether or not they went on to develop liver cancer following an adult diagnosis of T2DM. They incorporated 10 variables into a model of cancer risk (gender, age, alcoholic cirrhosis, nonalcoholic cirrhosis, alcoholic hepatitis, viral hepatitis, other types of chronic hepatitis, alcoholic fatty liver disease, other types of fatty liver disease, and hyperlipidemia) and subsequently constructed artificial neural network (ANN) and LR predictive modeling approaches. The authors found significantly superior classification accuracy using the ANN approach and have now moved on to the next stage, which is construction of a web-based application for ML directed prediction of liver cancer risk in diabetic patients using the described ANN algorithm.

Although these studies illustrate the potential for AI-based approaches in cancer risk stratification, both examples involved analysis of well maintained, prospectively curated databases. The key challenge in real-world clinical environments is that patient data is all too often anything but well maintained and complete, and accessing and collating this data in a meaningful way to extract a quantifiable “risk” of cancer for patients is challenging as a direct consequence. A recent collaborative venture between clinicians at MD Anderson Cancer Center and IBM Watson Health sought to develop an AI application (Oncology Expert Advisor, OEA) capable of providing real-time, real-world patient-specific decision support [15]. This study was primarily borne out of an acknowledgment of the disparate nature of patient health records. The collaborative group, comprised of clinicians and computational scientists, set about creating an AI solution equipped with an advanced natural language processing (NLP) algorithm capable of automatically locating, extracting, and analyzing patient data to compile a composite patient-specific profile, which in turn can be used to drive clinical decision making. The authors found the process of applying NLP to medical records far more challenging than applying AI-based approaches to the evaluation of the published literature or publicly available data repositories. This was due to the fact that medical record documentation frequently comprises fragmented sentence structure, grammatical errors, acronyms, and statements of ambiguous meaning. The developers attempted to overcome these problems by training the OEA to recognize data specifically across nine “concept” modules. Through this approach, the OEA application was found to be capable of locating, searching, and extracting relevant endpoints from medical records, and using this data to create an “integrated patient profile” comprised of multidimensional clinicopathological data. The profile for a given patient, as depicted by OEA, is continually updated, as new data is inputted. The authors assessed the performance of OEA for searching disparate sources within electronic medical records and extracting complex clinical concepts from unstructured text documents varied, reporting F1 scores of 90%–96%. Based on OEA-based patient profile constructs, approved therapies linked to the available supporting evidence were identified with 99.9% recall and 88% precision, and patients were accurately screened for eligibility for clinical trial enrollment (97.9% recall; 96.9% precision). These and other emerging results demonstrate the technical feasibility of an AI-powered solution with which to construct context-specific longitudinal patient profiles with which to propose evidence-based treatment and/or trial options. The authors highlight the critical need for collaboration across clinical and AI domains, and the requirement for clinical expertise throughout all steps of the process, from study inception to design, training, and clinical validation. The developers have yet to test the platform outside of MD Anderson Cancer Center, but undoubtedly this represents an important development and highlights the potential for AI approaches to circumvent some of the fundamental challenges clinicians face with data mining.

5.4 Artificial intelligence for enhanced cancer diagnosis and staging

Clinical vignette: Penny is a 40-year-old telecommunications executive who has recently been diagnosed with breast cancer. Her surgeon counsels her about the treatment options and it is recommended that she undergoes a sentinel lymph node biopsy (SLNB) before deciding whether or not to proceed to more radical surgical axillary nodal clearance. Biopsy tissue acquired at the time of SLNB is sent to the histopathologist who subjects the slides to hematoxylin and eosin (H&E) staining, as well as immunohistochemical analysis. The verdict based on pathologist evaluation is that her nodes are free of cancer invasion. However, Penny’s cancer physician has also enrolled her to a new trial evaluating the application of deep learning algorithms in detecting lymph node metastases in women with breast cancer. On AI-

based analysis, her tissue sections are computationally considered to be cancer-bearing. She decides to undergo the more radical treatment option and currently remains in remission.

The above case serves to illustrate the challenges facing conventional immunohistochemical evaluation of biopsy samples in cancer medicine, where the stakes are high, and where a verdict one way or another, can be the difference between major surgical intervention and more conservative treatment. The patient in the case example elected, on the advice of her treating physician, to undergo more extensive surgical treatment but felt that this decision was offered credence by the fact that the DL algorithm had identified her lymph node as being cancer-bearing. A study published in JAMA in 2017 by Bejnordi et al., on behalf of the CAMELYON16 Consortium, sought to evaluate this very clinical scenario, by assessing the discriminative accuracy of DL algorithms in detecting lymph node metastases in tissue sections of women with breast cancer, compared with conventional histopathological interpretation [10]. This study is particularly novel since the principal investigators organized a competition (Cancer Metastases in Lymph Nodes Challenge 2016, CAMELYON16) inviting research groups around the world to produce an automated solution for breast cancer metastases detection in sentinel lymph nodes (SLNs). The intention was that once developed, the performance of each algorithm would be compared to that of a panel of 11 breast cancer pathologists. Over a 1-year period, 390 research teams signed up for the challenge, demonstrating the tremendous enthusiasm in computer science for the development of cancer-directed AI solutions. Of these initial declarations of interest, 23 teams submitted 32 algorithms by the competition closing date. In terms of methodology, slides corresponding to SLN tissue from 339 patients were included in the analysis. The top performing algorithm by team HMS and MIT II used GoogLeNet architecture and outperformed all other competition entrants with area under the curve (AUC) of 0.994 (95% CI, 0.983–0.999). This AUC demonstrated statistically significant superiority over mean pathologist performance (AUC of 0.810; 95% CI, 0.738–0.884; $P < .001$). Although this is an early proof of concept study, the findings would appear to suggest that the potential for computationally augmented cancer diagnostics seems indisputable, and moreover, the potential for these approaches to surpass human capabilities seems genuinely possible.

There are also ways where AI approaches can represent not only precise, but also frugally innovative solutions to current cancer challenges. A study published in Nature in 2017 by Esteva et al. acknowledged that skin cancer, representing the most common human malignancy, is primarily diagnosed based on visual evaluation of the lesion(s) in question, commencing with initial clinical examination in the doctor's office, followed potentially by dermatoscopic evaluation, biopsy, and histopathological assessment [16]. The authors developed and implemented a deep CNN in the evaluation of 129,450 clinical images, including 2032 different skin conditions. The authors compared classification accuracy of the CNN with the performance of 21 board certified dermatologists and found the CNN to demonstrate equivalent skin cancer classification accuracy compared with dermatologists. The really telling point here, as the authors point out, is that mobile devices outfitted with deep neural network capabilities would have the potential to greatly extend the “reach” of dermatologists trying to diagnose skin cancers at the early stage, outside of over-subscribed clinics. The potential here to provide a low cost, universally available solution for diagnosing such a common cancer using an AI approach is tremendous.

5.5 Artificial intelligence to predict cancer treatment response

Clinical vignette: Tomas is a 58-year-old university lecturer from Stockholm who has recently been diagnosed with a locally advanced rectal cancer. Preoperative imaging shows no evidence of distant site metastases, but pelvic MRI reveals that the primary tumor is bulky and extends to the predicted margin of surgical resection. As per conventional practice, it is suggested that Tomas undergoes pelvic radiotherapy, with a view to down-sizing the primary tumor and improving the likelihood of achieving complete surgical tumor clearance, with cancer-free margins of excision. Tomas receives 50.4 Gy of pelvic radiotherapy in divided fractions and then undergoes repeat imaging 10 weeks later to assess tumor response. Unfortunately, his tumor has shown no signs of meaningful regression and now appears to have spread to the liver and lungs. Tomas is bitterly disappointed and wants to know why he was put forward for radiation therapy if it was not going to work?

Typical questions in radiation oncology include “what is my risk of treatment-related toxicity?”, “what is the likelihood of achieving a good local response?”, and “will my tumor spread elsewhere while I am being treated?”. Unfortunately, the molecular and tumor-specific mechanisms that govern and regulate radiation response are not completely understood at the present time. However, increasingly, patients are wanting to be presented with a factually accurate projection of what radiation therapy will mean for them. AI approaches may represent an attractive way of generating a credible answer to this question and are beginning to demonstrate potential in this area. The processes of target and normal adjacent tissue image segmentation, treatment planning, dose optimization, and toxicity estimation are all areas that could benefit from AI solutions. It is possible in the future, with advances in AI to envisage a system that precisely identifies target and normal adjacent tissue volumes, estimates optimal radiation modality and beam arrangement, and determines a strategy that maximizes local tumor control, while minimizing toxicity. The AI approach should ideally have integrated access to electronic health records comprised of prospectively curated clinicopathological data as well as molecular data. The issue of how we realize the potential of AI-based approaches in the field was the subject of a recent review by Thompson et al. which explores some interesting themes [17]. The authors review the potential for AI as a positively transformative element in radiation oncology and provide a detailed discussion on key challenges in realizing this potential in terms of data availability, data equity, education and training and funding. They suggest that the international radiation oncology community will need to work collaboratively with industry and academia, to ensure optimal utilization of investment, resources, technology, and personnel.

Achieving the goal of precision radiation oncology using AI will be contingent on high-quality research in the field. Toratani et al. published a study in Cancer Research in 2018 where they employed a CNN to evaluate cancer tissue sections and found that this approach was capable of distinguishing cancer regions likely to be radiation sensitive versus radiation resistant, based on the manner in which they were seen to cluster, on computational image segmentation [11]. The ability to use such a technique at the time of diagnosis and treatment planning will potentially allow patients who are predicted to have radiosensitive tumors to be put forward for radiotherapy, while others may need to go directly to surgical resection, as the unnecessary delay to definitive treatment (surgery) while undergoing ineffective local treatment, risks distant cancer spread, as pointed out in the illustrative vignette provided above.

As with radiation therapy, chemotherapy works effectively in a proportion of patients and in the precision oncology era, it is essential to be able to reliably select the right drug for the right patient to maximize anticancer drug efficacy, ideally with minimal toxicity. Drug resistance, a phenomenon that describes tumor evasion of anticancer drug actions, is considered to represent the primary reason for

chemotherapeutic treatment failure. The implication here is that a small cohort of cancer cells either acquire (acquired drug resistance) or innately feature (primary drug resistance) the ability to become “immune” to therapy, via multiple, complex drug-resistance mechanisms. A key challenge in drug resistance is the development of methods with which to predict, ideally in advance of treatment initiation, the likelihood of drug effectiveness/resistance. At the present time, monitoring treatment response typically involves metabolic imaging using positron emission tomography. However, this is usually only undertaken 1–2 cycles into the course of treatment, which in recurrent cases risks cancer progression and unnecessary toxicity. Given the unmet need in this area for enhanced response prediction, Liu et al. conducted a study to evaluate the use of single cell mass spectrometry combined with ML to provide a means of ultrafast prediction of treatment response [18]. Single cell MS metabolomic profiling was used to derive rich biochemical information, considered by the authors to be the summative endpoint of upstream genomic/proteomic/transcriptomic activity. Acquired data were then subjected to ML interrogation with the aim of identifying “hidden” metabolic patterns via clustering, regression, and predictive modeling. A variety of algorithms were assessed for predictive accuracy, and the ANN approach demonstrated overall superiority with over 95% predictive accuracy of drug-resistant cancer phenotypes. This work is particularly exciting for several reasons. First, it demonstrates the ability to explore cancer cellular heterogeneity effectively with the single cell MS technique. Second, the combination of minimal sample preparation combined with ML methods that provide rapid results led the authors to conclude that this approach could be used for point-of-care prognostic/therapeutic assays. An area where this would have game-changing potential for the cancer surgeon would be in the field of peritoneal surface malignancy. In operable cases, this is usually treated with a combination of radical cytoreductive surgery, followed by intraperitoneal instillation of hyperthermic chemotherapeutic agents, in liquid form. At the present time, there is no effective means of defining which intraperitoneal chemotherapeutic drug to use for a given patient. A method with which to acquire a sample, intraoperatively, perform single cell MS analysis and combine this with ML to select the most effective drug for a given patient would be highly desirable and shows how high-throughput bioanalytical approaches combined with AI will create the next generation of methods with which to not only select chemotherapeutic regimens, but also to inform the cancer surgeon in real time.

Other previous studies have also demonstrated the ability of AI approaches to predict treatment response. For example, Huang et al. recently looked at ML as a means of predicting individual cancer patient responses to chemotherapy [19]. The authors subjected gene expression profiles of 152 cancer patients to ML interrogation and found that the computational approach permitted highly accurate (>81%) prediction of likely response to either gemcitabine or 5-FU. The authors had earlier published work describing an open access support vector machine (SVM)-based algorithm with which they were able to predict the likely collective response of 273 ovarian cancer patients to seven commonly prescribed chemotherapeutic agents [20]. However, with this more recent study they demonstrated the ability of AI-based approaches to predict response at the individual level, using person-specific gene expression data.

5.6 Artificial intelligence to predict cancer recurrence and survival

Clinical vignette: *Feroz is a 62-year-old retired mechanical engineer who was diagnosed with colon cancer in 2011. His surgical oncologist subjected him to a full raft of investigations before offering surgery, which took the form of a laparoscopic right hemicolectomy. His resection specimen revealed a completely excised Stage I tumor of the ascending colon. His case was discussed at the colon cancer multidisci-*

plinary meeting, and given the favorable pathology, he was not felt to require adjuvant chemotherapy. He underwent yearly CT scanning and follow-up and was told by his physician in 2016, that having completed 5 years of follow-up, and having remained recurrence free, he would be discharged from further routine surveillance as this was not indicated. In May of 2018, Feroz sees his doctor for unintentional weight loss and a CT scan is arranged. This reveals a lesion in the right lung, which is subsequently confirmed on biopsy as a colon cancer metastasis. Feroz and his family are confused; they were told to complete 5 years of follow-up, which they did, and cannot understand why this has happened.

The vignette above illustrates the critical need to develop more personalized and more precise cancer surveillance pipelines with which to follow up cancer patients. The fundamental problem with conventional cancer surveillance approaches is that they are essentially designed to detect recurrence. Ideally, the objective should be to predict likelihood of recurrence and prevent this from happening if possible with appropriate treatment modifications. At the present time, this is not possible, though there are mounting efforts toward the use of AI-based approaches in predicting the likelihood of cancer recurrence, and these are beginning to show genuine potential.

Gastric cancer is a good case exemplar, as it is a leading cause of cancer-related mortality worldwide, associated with abbreviated 5-year survival, even in treated cases, due to a strong propensity for recurrence. A recent study by Zhang et al. used gastric cancer as a case model and the authors described identification of long noncoding RNAs (lnc-RNAs) involved in gastric cancer development, progression, and patterns of relapse [21]. A composite signature comprised of 11 lnc-RNAs was identified using a ML approach applied to gene expression profiles obtained from the National Center for Biotechnology Information (NCBI), Gene Expression Omnibus (GEO), ArrayExpress, and TCGA data repositories. A weighted correlation network analysis algorithm was constructed to permit ML-based mining of gastric cancer-related molecular data. A “risk-score” was calculated for each patient based on expression patterns of the 11 lnc-RNAs, and patients were classified as “low-risk” or “high-risk,” accordingly. The authors demonstrated that ML derived risk scores correlated strongly with overall and recurrence-free survival in gastric cancer patients. They suggest that this technique could offer a means of stratifying patients more individually and planning their treatment and subsequent surveillance in more personalized fashion.

Cancer surveillance is fundamentally intricate. We tend to survey patients for largely arbitrarily defined periods of time following their cancer treatment in completely generic fashion. The problem here is that the inherent longitudinal complexity of what follows once a patient has had cancer treatment means that the “one size fits all” approach in surveillance is severely limited. Ideally, a method for precisely surveying the cancer patient comes down to factors encompassing the past, the present, and the future, and computationally these are difficult to model. There has been interest in the application of next-generation DL approaches in order to longitudinally map the cancer patient “trajectory” post treatment using NLP architecture. These approaches currently remain in the early phases of development but are demonstrating strong potential. It is again important to acknowledge here the unavoidable computational workload that comes with models of this sort that are trained by ever larger and larger datasets. An elegant example of how AI can be successfully applied in cancer surveillance, while simultaneously mitigating problems of computational over-burden is provided by a study published by Qui et al. who looked at distributing the computational workload required for such a task across multiple computing cluster nodes, using a state-of-the-art data parallelism algorithm [22]. A high-performance computing environment was used where a Gradient Descent data parallelism algorithm was

used to train a CNN for a NLP algorithm that involved extracting and modeling information from a large-scale cancer database. The authors reported encouraging results and concluded that this approach could represent a feasible means of defining surveillance pathways for cancer patients in the future.

In another study evaluating prediction of oral squamous cell carcinoma (OSCC) recurrence patterns, Exarchos et al. proposed a multiparametric decision support system in order to investigate the basis for OSCC relapse after seemingly total remission in patients undergoing radical treatment [23]. They modeled multiple heterogeneous sources of data (demographic, pathological, imaging, and genomic) in order to predict the likelihood of OSCC recurrence from a cohort of 86 patients, of whom 13 succumbed to loco-regional recurrence. A variety of ML algorithms were used to interrogate the data and among the evaluated variables, an AI generated a composite “panel” of markers consisting of smoking history, tumor thickness, radiological evidence of extraoral tumor spread, number of enlarged lymph glands seen on CT imaging, and mutational status of a panel of genes comprising p53, SOD2, TCAM, and OXCT2 genes, allowed robust discrimination of patients into those with a strong likelihood of tumor recurrence and those predicted to remain disease-free.

Park et al. reported on the use of ML techniques for the analysis of breast cancer recurrence [24]. The proposed algorithm is based on the construction of a graph model depicting recurrence risk which integrates gene expression data with gene network interactions. Based on biological knowledge, the authors selected gene pairs that indicate strong biological interactions. Patient samples were classified into three groups: (1) recurrence, (2) nonrecurrence, and (3) unlabeled samples. Using a combined total of 108,544 gene-encoded protein network interactions, the authors were able to demonstrate that a semisupervised ML method was capable of identifying a number of critical genes related to cancer recurrence. They also claimed that this approach was found to outperform other existing methods suggested for breast cancer recurrence prediction.

A recent study by Hasnain et al. from the University of Southern California evaluated the role of ML approaches in predicting recurrence patterns and survival outcomes in a cohort of 3503 patients undergoing cystectomy for bladder cancer [25]. The authors constructed initial multivariate models using a series of “base” algorithmic models consisting of SVM, bagged SVM, K-nearest neighbor (KNN), adaptive boosted trees (AdaBoost), random forest (RF), and gradient boosted trees (GBT). These were then used to create more sophisticated “ensemble” models incorporating multiple algorithms simultaneously along different phases of the data treatment pipeline. ML methods developed a meta-classifier comprised of clinical and pathological data that predicted 5-year recurrence with sensitivity and specificity of 70% and 70%, respectively. Using the same approach, 5-year survival was predicted with sensitivity and specificity of 74% and 77%, respectively.

5.7 Artificial intelligence for personalized cancer pharmacotherapy

The potential for AI as a powerful positive disruptor in cancer pharmacotherapy is tremendous. In broad terms, the application(s) are likely to fall under three main headings: (1) *drug selection and toxicity prediction*; (2) *drug pairing*; and (3) *drug repurposing*.

In terms of drug selection, [Section 5.5](#) has highlighted the irrefutable potential for AI-based approaches to determine whether a given patient will respond favorably to a particular anticancer treatment. On the flip side of this point lies toxicity prediction, where the case is equally clear regarding the need for improved pre-

diction of unwanted treatment-related side effects; to put this need into perspective, recently published figures from the National Cancer Institute (NCI) estimate that there are currently 16.9 million cancer survivors in the United States, a figure set to rise to a projected 26 million by 2040 [26]. A significant proportion will receive chemotherapy in the course of treatment. Unfortunately, approximately 50% of these patients will experience drug-related side effects, including cardiovascular, dermatological, gastrointestinal, and neurological [26]. Neurotoxicity is common and chemotherapy-induced peripheral neuropathy, during and after treatment, frequently with permanent and debilitating consequences, has been reported to occur in an estimated 30%–40% of patients receiving chemotherapy. Bloomingdale and Mager recently reported the use of a ML method combined with drug structural analysis to computationally model the likelihood of drug-induced neuro-toxicity (peripheral neuropathy) from a manually curated library of 95 approved chemotherapies [27]. Molecular descriptors were generated for each of the 95 drugs. MDL MOL files for each drug were obtained from public databases (PubChem, DrugBank, and ChEMBL). MDL MOL files were imported into ADMET Predictor, and 227 molecular descriptors were generated, which included descriptors from the following categories: electrotopological states, topological indices, charge based, functional groups, hydrogen bonding, simple constitutional, and ionization. ANN models were developed for the quantitative prediction of peripheral neuropathy incidence and SVM models were subsequently developed to classify drug compounds into three groups (high, medium, or low) according to their propensity to lead to peripheral neuropathy. Using various algorithmic approaches, they reported sensitivity and specificity of 82%–90% and 87%–97%, respectively. They were able to isolate the number of aromatic rings in the molecular structure of evaluated drugs as a frequent and sensitive factor associated with propensity for neurotoxicity. What was particularly impressive with this study was the fact that the authors then applied the same ML approach to predicting the likely incidence of peripheral neuropathy for 60 small molecular drug compounds currently under clinical trial investigation. This has enormous potential implication for anticancer drug development, as increasingly perceived tumoricidal benefits will need to be weighed up against likely treatment-induced toxicity, and up to this point, it has not been possible to reliably quantify the latter for most drugs.

Drug pairing is another area where AI-based methods are likely to feature strongly moving forward. Cancer is highly complex with a single tumor thought to develop on the order of 100 million coding region mutations, with this molecular diversity believed to be a fundamental cause for drug resistance and treatment failure. Targeting multiple areas of a given tumor's molecular infrastructure requires both a detailed understanding of intratumor biomolecular interactions and also an understanding of how multiple drugs if used in combination will interact with one another. In this regard, the concept of drug pairing in cancer treatment is generating considerable enthusiasm. The fundamental complexity of this endeavor has proved prohibitive until recently, but AI-based approaches are starting to have a big impact here. An elegant example of this was provided recently by Xia et al. who applied a DL module to evaluating and predicting tumor cell-line responses to single-agent therapy versus combinations of drugs selected from 104 FDA-approved anticancer drug compounds, using the recently published NCI ALMANAC resource [28]. The authors modeled the combined activity of anticancer drugs and noted good model performance ($R^2=0.94$) concluding that the methods employed represent a promising means of selecting drug combinations with greater anticancer efficacy compared with single-agent treatment. The DL algorithm considers tumor molecular phenotypic information including gene expression, microRNA expression, and protein expression as well as drug compound structural fingerprints to create the optimal therapeutic “match” for a given tumor cell line and the best performing combination treatment. This approach has the potential to revolutionize the way cancer patients are treated in

the future—one can imagine acquiring a biopsy sample from a given tumor, subjecting it to organoid culture and molecular phenotypic interrogation, before using an AI algorithm to predict the most effective treatment and validate this in vitro.

Drug repurposing, or repositioning, represents an area of rapidly growing interest in cancer, with the potential to reduce the cost of new drug development by applying repurposed existing drugs on novel drug–target networks. The “one gene, one drug, one cancer” paradigm is now being challenged and AI based are at the forefront of network-based algorithms now in development that explore drug–drug similarities, drug–target relationships, and gene–gene interactions. An early example of this potential was presented by Iorio et al. who demonstrated that drugs exhibiting chemical structural commonality and similar transcriptional response profiles can potentially be repositioned with one another for use in, at times, highly disparate clinical pharmacological indications [29]. The authors predicted similarity in terms of the drug therapeutic effect by comparing gene expression profiles following drug treatment across multiple cell lines. An alternative way of using AI to repurpose drugs evaluates drugs and targets as part of a wider cancer molecular network. For example, Cheng et al. used a ML approach to predict drug–target relations based on a network of three distinct types of network connectivity: (1) drug–drug structural commonality, (2) target–target similarity, and (3) drug–target/drug–target interaction commonality [30]. The authors performed experimental validation of proposed drug repositioning strategies, for example, demonstrating that simvastatin, a drug used to lower cholesterol, through exertion of an acknowledged inhibitory action on HMG-CoA reductase, also demonstrated powerful antagonistic effects on estrogen receptor β which is thought to be involved in breast cancer development and progression. On validatory in vitro work, the authors found simvastatin to exert a moderate, novel antiproliferative effect on breast cancer cell lines.

5.8 How will artificial intelligence affect ethical practices and patients?

Fostering public and commercial enthusiasm for AI in cancer medicine is critical to the success of the precision oncology endeavor. From the public perspective, there is interest, but this is also perhaps tempered by a degree of skepticism. One of the issues not discussed in the previous sections relates to the fundamentals of the doctor–patient axis. The rise of AI and the dawn of computationally directed decision making for the cancer patient will fundamentally challenge the age-old relationship between patient and treating physician. In some instances, the patient is likely to perceive this change in dynamic as a beneficial one—for example, the drug carefully selected, via AI, to minimize side effects and maximize treatment response. However, what if these algorithmic approaches are introduced to the cancer multidisciplinary board, as a “virtual” member? This is on the horizon, and recent studies have been seeking to evaluate the potential of “Intelligent Systems Support” for multidisciplinary medical teams [31]. Medicine has at its core a humanistic foundation—decisions on treatment for cancer are reached through consensus opinion of the cancer board at most units, taking into account local experience, local capabilities, and the available evidence. When a team of human physicians, including experts from multiple specialties, arrive at a conclusion (e.g., to offer the patient curative intent surgery, or to accept that the situation is too advanced and instead offer the patient palliative therapy) this news is discussed with the patient and often makes for an emotional and difficult encounter for all parties. But it is one human physician, discussing and determining the right treatment, with a human patient; and the encounter benefits from the centrally “human” nature of this. An AI colleague in the cancer multidisciplinary team (MDT) is probably an inevitably going forward, but it will present an addi-

tional layer of complexity to these difficult clinical situations. An AI platform is likely to be able to predict the patient's anticipated gains with treatment in terms of prolonging survival, estimate likely costs to the institution and estimate costs to society as a whole, and will come up with an answer of its own with unparalleled precision. But is precision really the panacea in such a situation? Many patients with cancer when faced with slim prospects for long-term cure still want to give radical treatment a go. It is the only chance they have; and yet AI will dispense with sentiment and instead will suggest proceeding with costly and risky treatment only when it is computationally felt to be feasible/advantageous. If we introduce state-of-the-art AI into the cancer MDT, what happens where there is fundamental disagreement between human board members and AI? There are no right or wrong answers with these questions currently, but what is clear is that AI approaches to cancer medicine will have social and legal implications that will need to be thoroughly explored in the future. From the point of view of industrial buy-in, the world would undoubtedly be better off without cancer, but how would this affect the pharmaceutical industry and is big pharma ready for cancer to be cured? Here, it is important to have an appreciation of spending; in 2015 the estimated spending on cancer medicines was around \$107 billion worldwide, according to data from IMS Institute for Healthcare Informatics [32]. This figure is expected to increase further to around \$150 billion by the year 2020. Although AI could have a highly disruptive effect on drug development and sales, this need not have a negative impact on the pharmaceutical industry. For example, recent analysis by McKinsey suggests that big pharma companies could generate in excess of \$100 billion using AI annually through research and development (R&D) alone [33]. The potential is therefore clear, and all stakeholders stand to make significant gains, provided AI approaches are introduced into mainstream cancer medicine with foresight and careful direction. Consortia made up of international experts in cancer medicine, regulatory organizations, academia, industry, computational scientists, and patient groups will need to steer the cancer AI venture to ensure that it lives up to the promise and has the positive impact that is hoped for.

5.9 Concluding remarks

AI-based approaches comprise a broad range of rapidly expanding and increasingly sophisticated computational methods which have the potential to act as major disruptive influences across all phases of the cancer patient pathway, from prediction of cancer risk, through to next-generation diagnostics, targeted therapy, and personalized cancer surveillance (Fig. 5.4). Although the currently available literature demonstrates the undoubted potential of these techniques, there are vast differences between studies in terms of the specific areas of application and the algorithmic method(s) employed. International consortia comprised of experts in clinical oncology, computational science, and translational research are required to try and define a more unified strategy moving forward and to outline an initial set of realistic deliverables, which can be subjected to large-scale, multinational randomized controlled trial evaluation. In parallel, the same challenges that broadly apply to the successful implementation of precision medicine approaches, apply here: pan-stakeholder collaboration, cost-effectiveness and quality assurance and high-quality research to confirm the benefits of any novel proposed approaches [34].

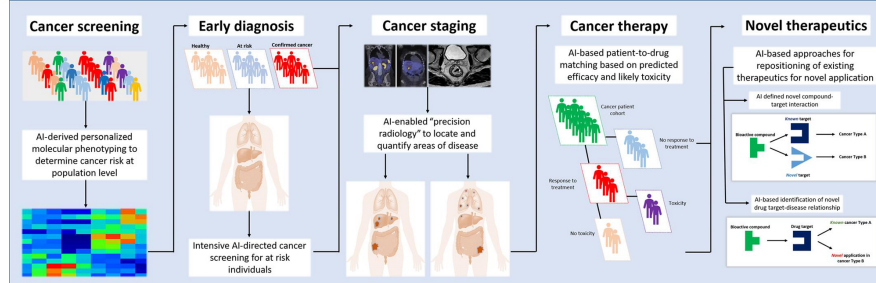


FIGURE 5.4 The application of AI-based approaches at all stages of the clinical cancer care pathway, from enhanced population-wide screening, to development of individual-specific pharmacotherapeutic regimens.

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